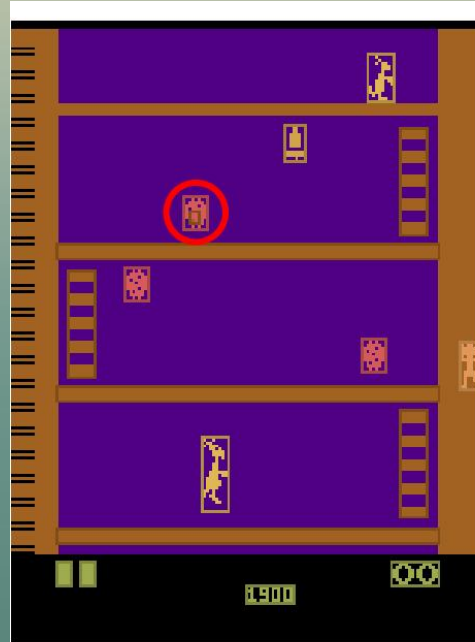


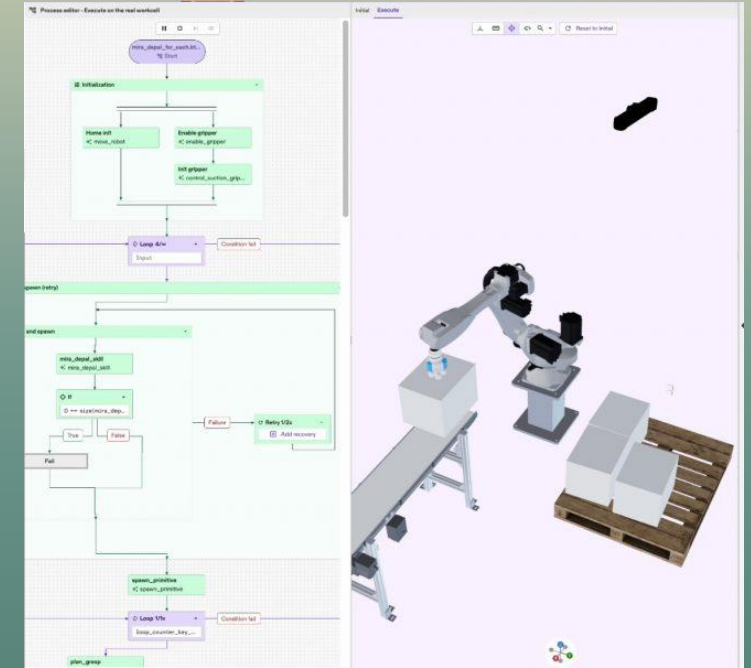
# GOSIM Paris 2026

May 5-6, 2026 Station F, Paris





Bell at (93, 36), (6, 11),  
 Child at (121, 12), (8, 15),  
 Enemy at (152, 109), (6, 15),  
 Fruit at (119, 108), (7, 10),  
 Fruit at (39, 84), (7, 10),  
**Fruit at (59, 60), (7, 10),**  
 Life at (16, 183), (4, 7),  
 Life at (24, 183), (4, 7),  
 Platform at (16, 124), (128, 4),  
 Platform at (16, 172), (128, 4),  
 Platform at (16, 76), (128, 4),  
 Player at (65, 141), (8, 24),  
**Projectile at (61, 65), (2, 3),**  
 Scale at (132, 132), (8, 35),  
 Scale at (132, 37), (8, 35),  
 Scale at (20, 85), (8, 35),  
 Score at (129, 183), (15, 7),  
 Time at (80, 191), (15, 5)]



PhD in symbolic RL agents  
 Research scientist - Google Intrinsic



# CONTENTS

GOSIM Paris 2026

## Part 01

(M)/RL agents learn shortcuts

## Part 02

We need structure + inductive biases

## Part 03

LLMs to generate symbolic policies

## Part 04

VC-dimension



# DQN on Pong



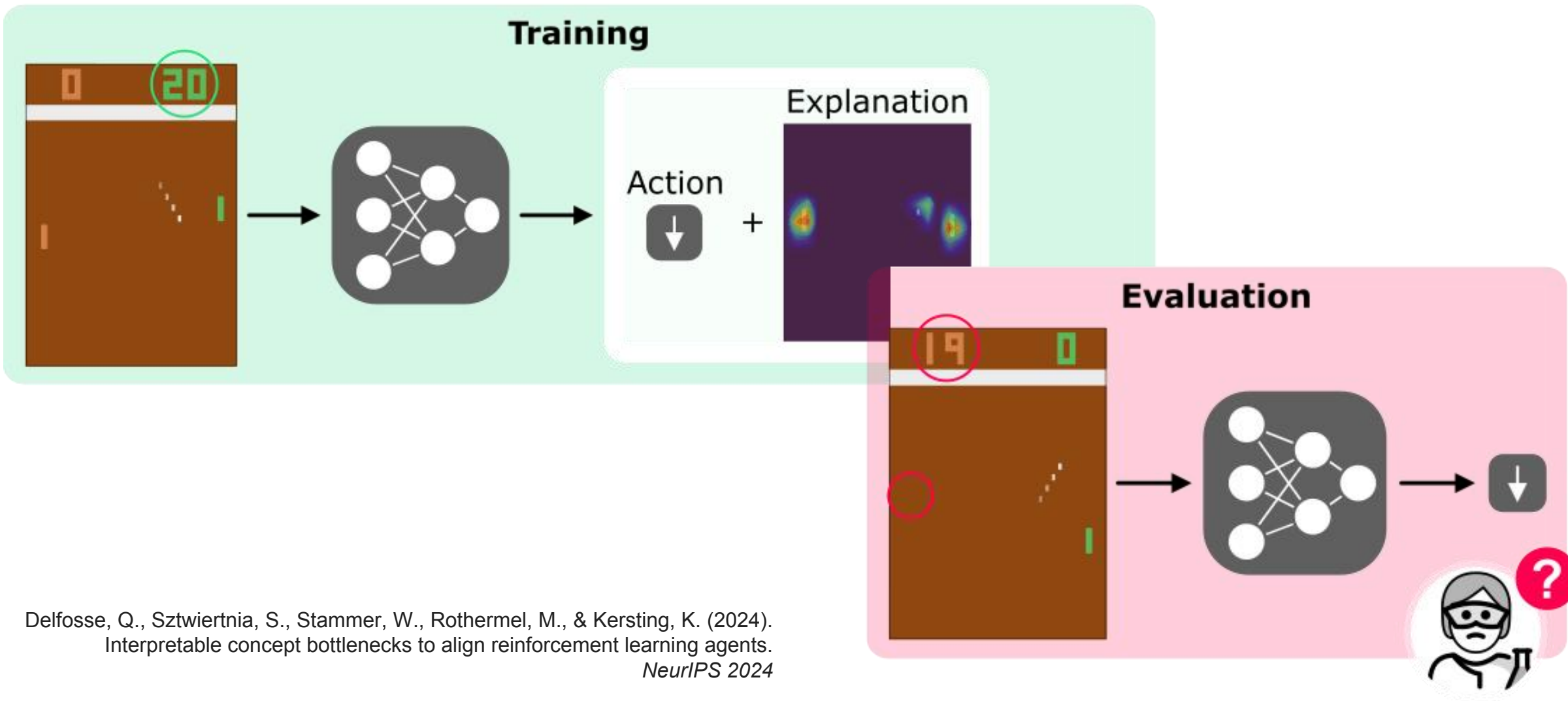
Mnih, V., Kavukcuoglu, K., Silver, D., Graves, A., Antonoglou, I., Wierstra, D., & Riedmiller, M. Playing atari with deep reinforcement learning. *arXiv preprint*. (2013).

Mnih, V., Kavukcuoglu, K., Silver, D., Rusu, A. A., Veness, J., Bellemare, M. G., ... & Hassabis, D. Human-level control through deep reinforcement learning. *nature*. (2015).



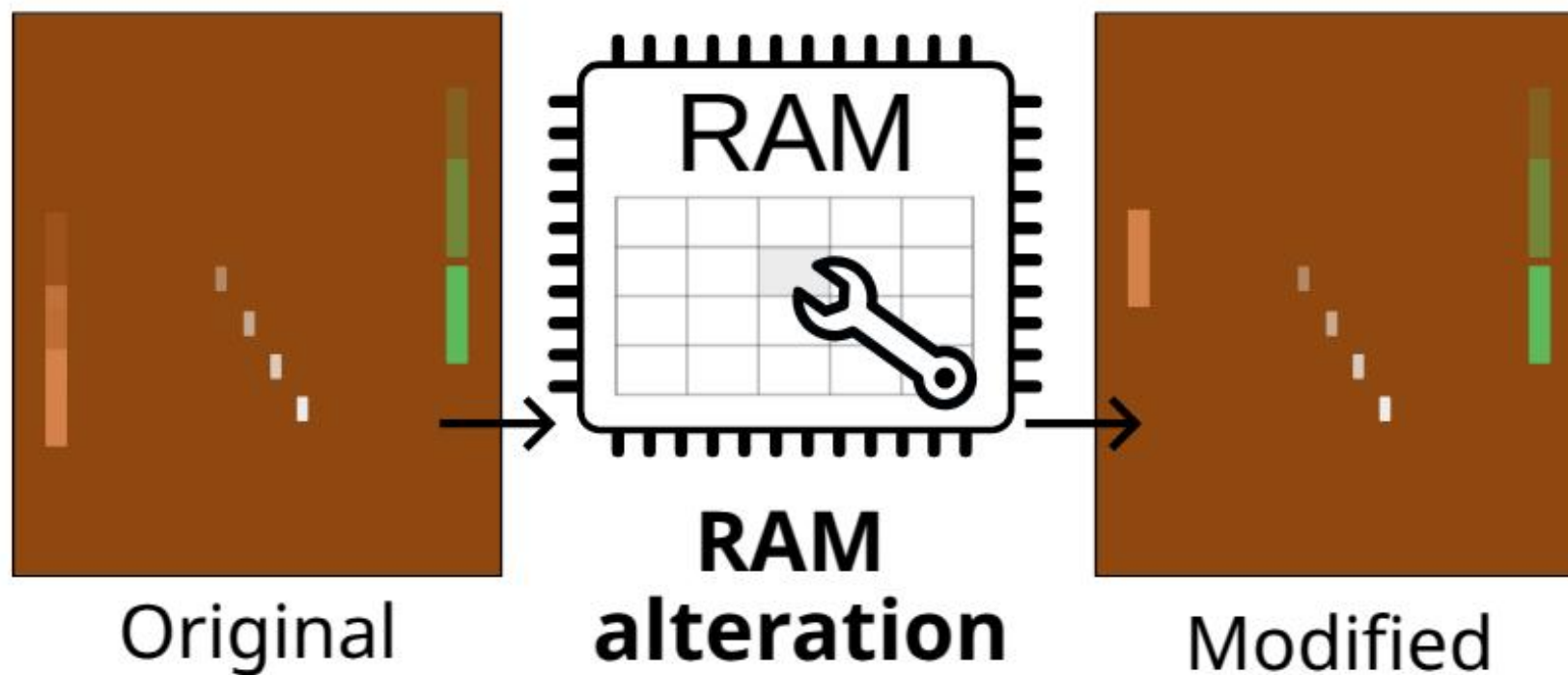
# Why interpretability?

RL agents work (and LLM know everything)



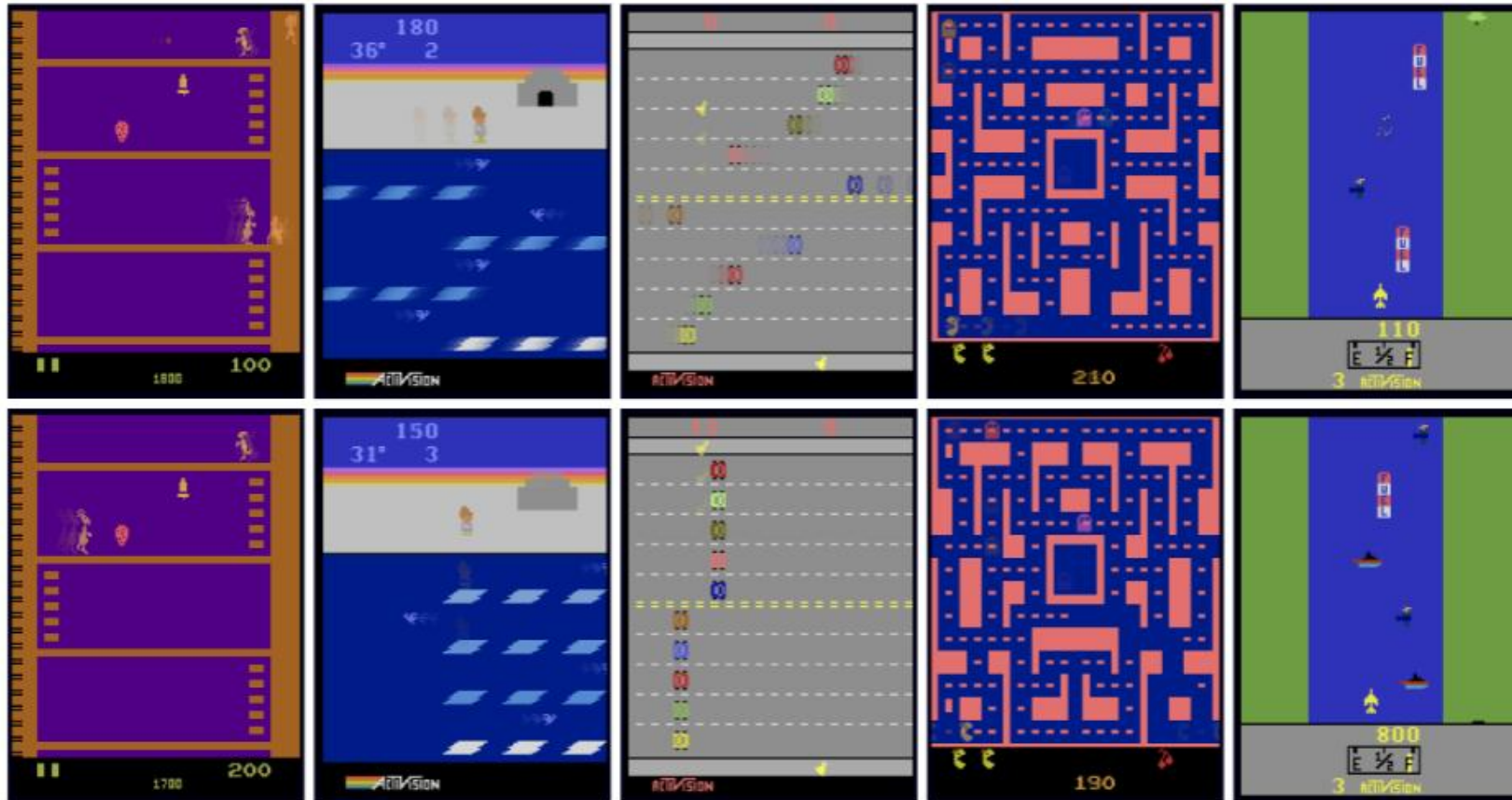
# Pong is an isolated case

Let's run a systematic evaluation – HackAtari



# Pong is an isolated case

Let's run a systematic evaluation – HackAtari



No Danger

Static Platforms

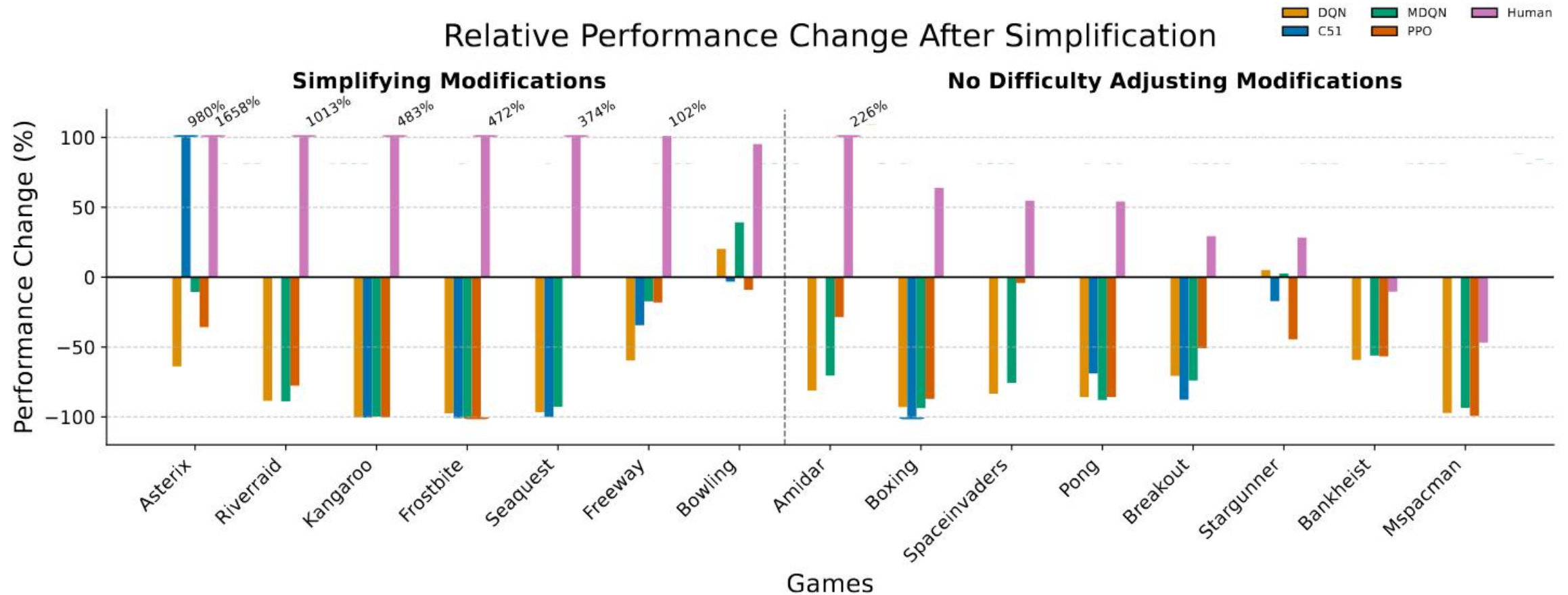
Stopped Cars

Change maze

No shot

Delfosse, Q., Blüml, J., Tatai, F., Vincent, F., Dillies, E., Gregori, B., K. Peters, J. & Kersting, K. (2024). Deep RL agents are not close to human intelligence.

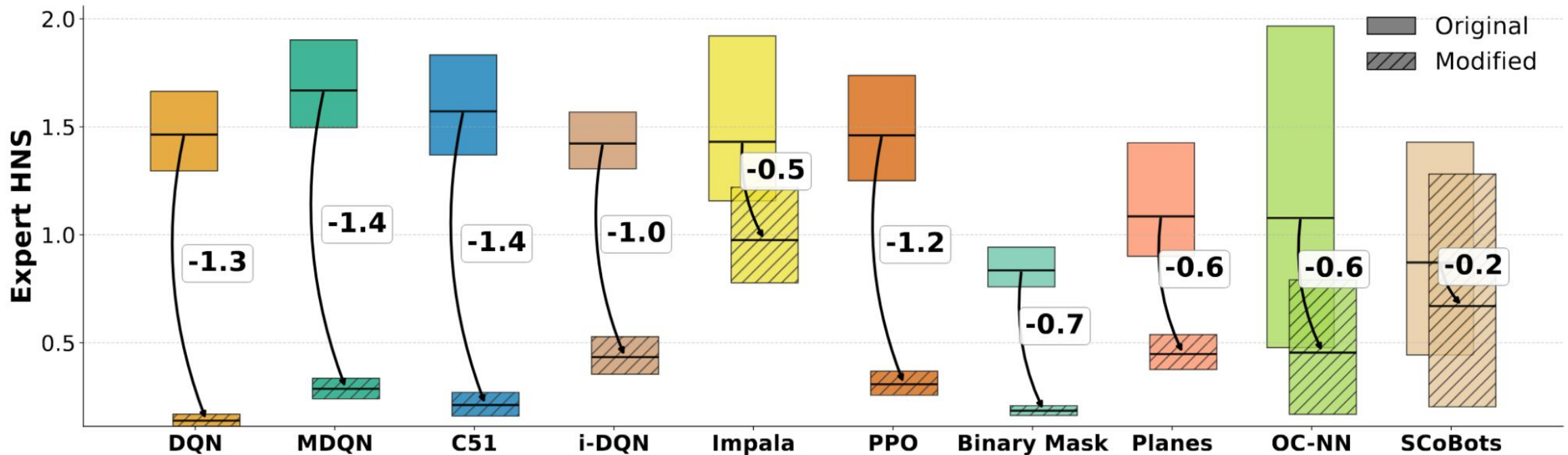
# RL agents are not close to human intelligence



Delfosse, Q., Blüml, J., Tatai, F., Vincent, F., Dillies, E., Gregori, B., K. Peters, J. & Kersting, K. (2024). Deep RL agents are not close to human intelligence.



# RL agents are not close to human intelligence



# It is not Atari-specific



De Haan, P., Jayaraman, D., & Levine, S.  
Causal confusion in imitation learning.  
*Neurips* (2019).

\*it did not happen to a Waymo



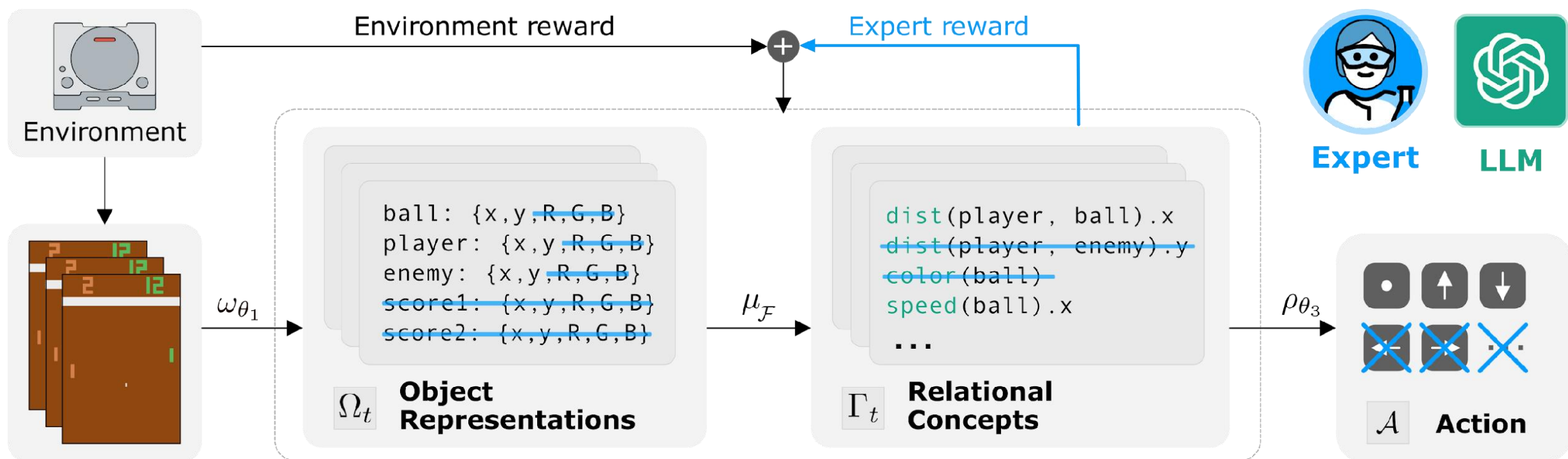
# Why are we tricked?



The Clever Hans effect

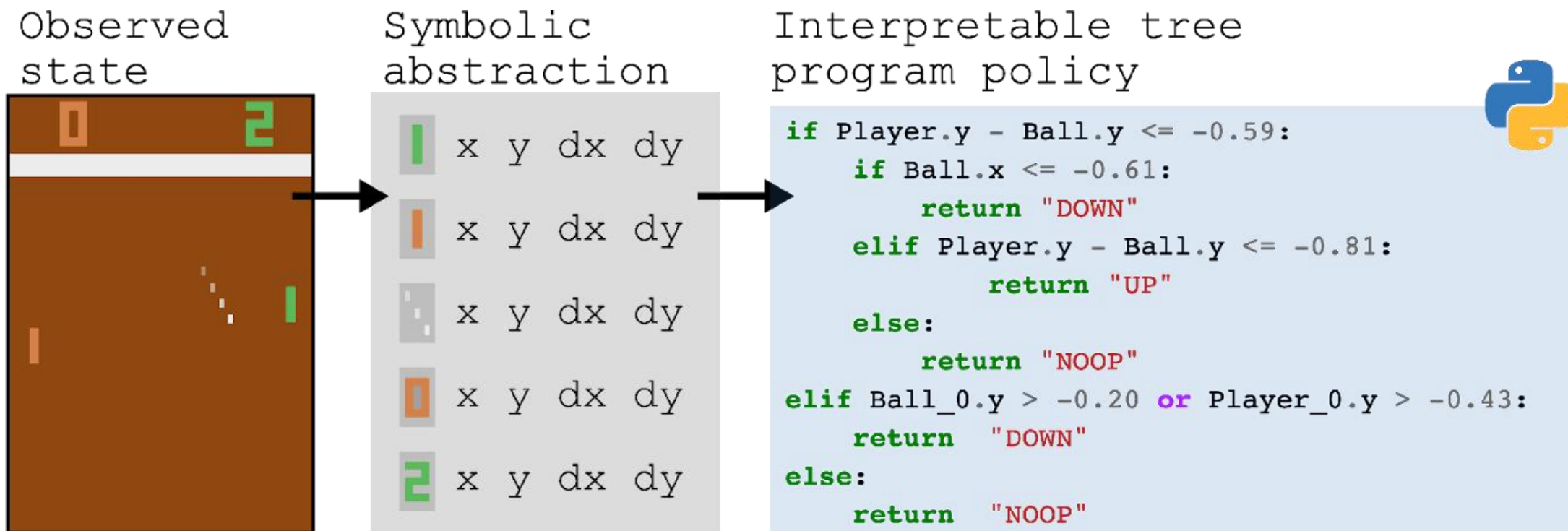
# What can we do ?

Force agents to use human concepts



# Transparent policies

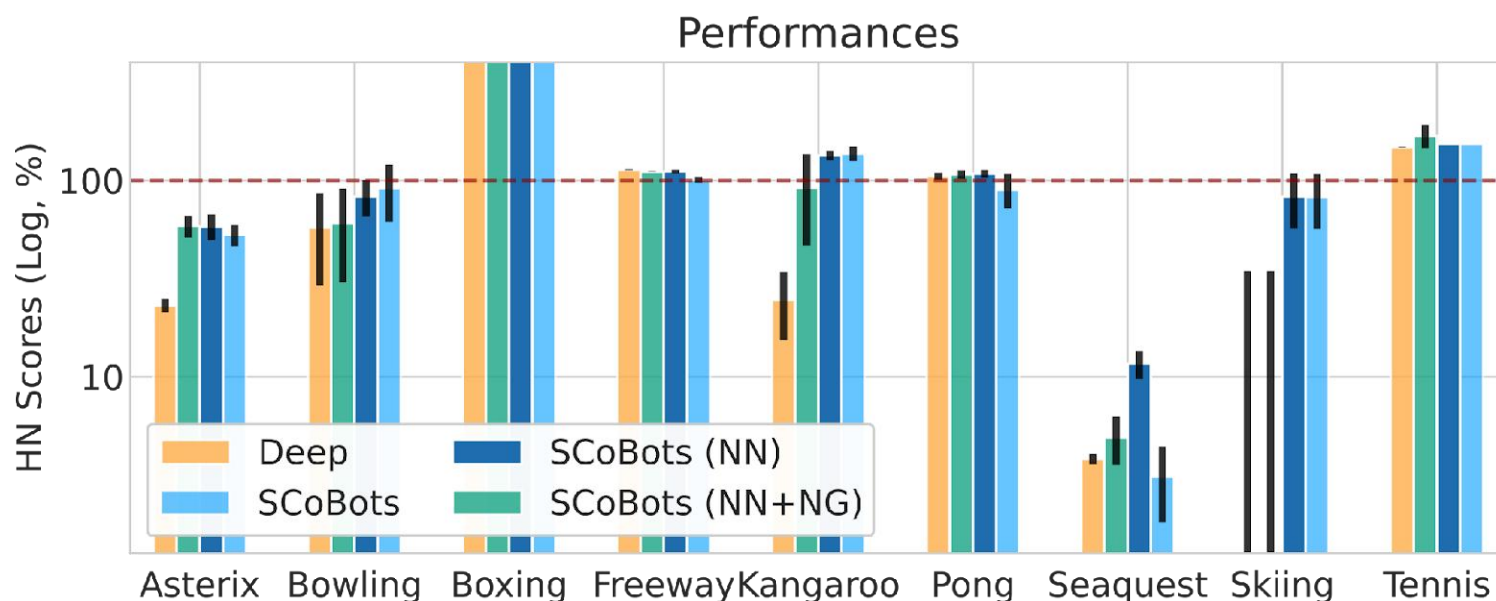
Use decision trees (or python programs)





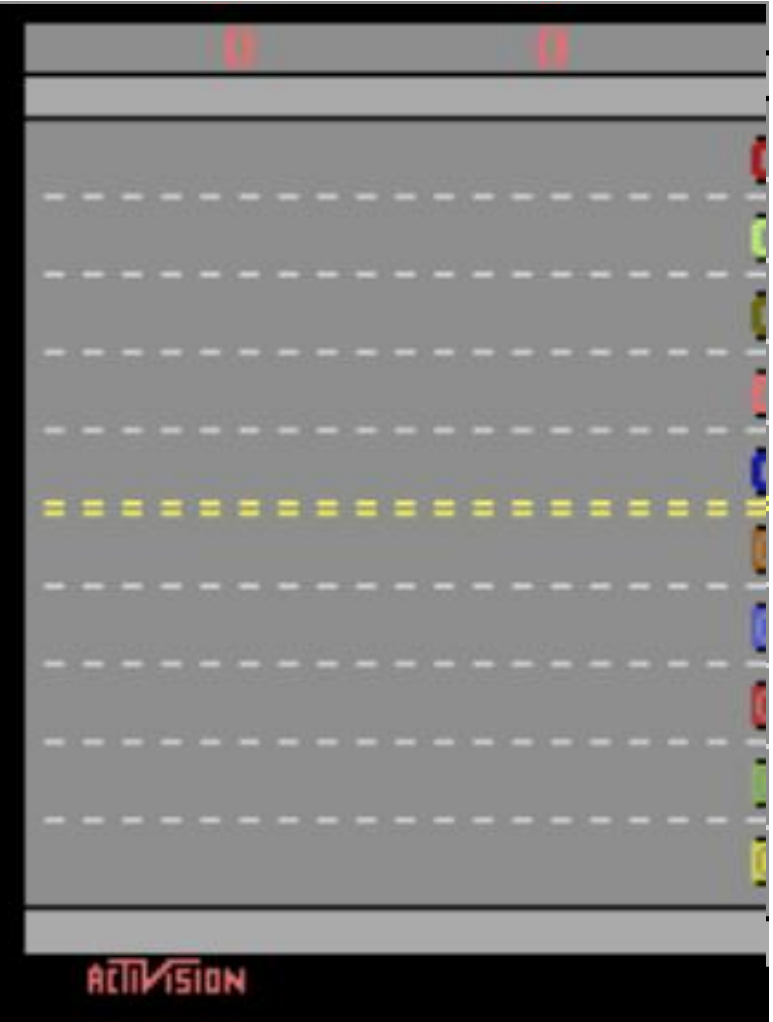
# What can we do ?

Extract and use human concepts



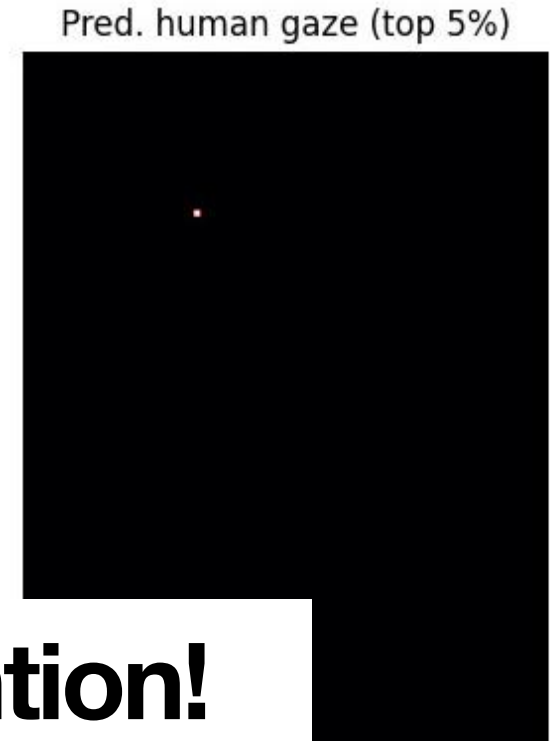
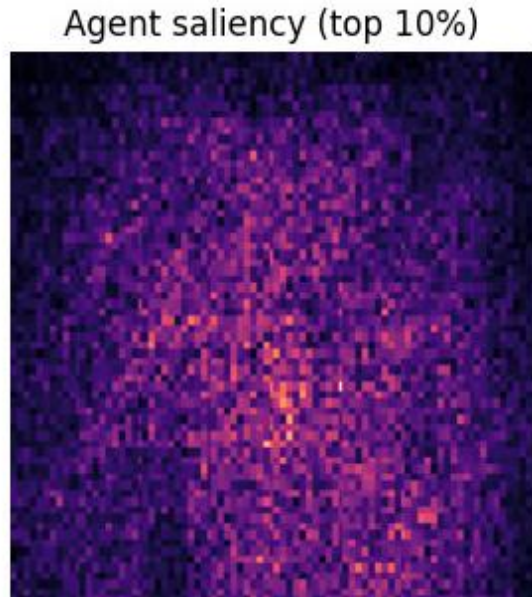
# Transparent trees are too deep

```
def play(state):
    if state.Chicken_0.y <= 1.22:
        if state.Chicken_0.y - state.Car_5.x <= 1.58:
            if state.Chicken_0.y - state.Chicken_0.prev_y <= -0.00:
                return "UP"
            else:
                if state.Chicken_0.y - state.Car_2.x <= 0.37:
                    return "UP"
                else:
                    if state.Car_6.prev_x - state.Car_8.prev_x <= -0.92:
                        if state.Chicken_0.prev_y - state.Car_4.prev_x <= 1.21:
                            if state.Chicken_0.prev_y - state.Car_4.x <= -0.16:
                                return "UP"
                            else:
                                return "NOOP"
                        else:
                            return "UP"
                    else:
                        if state.Car_2.x - state.Car_7.prev_x <= -0.01:
                            if state.Car_6.x - state.Car_8.x <= 1.32:
                                return "UP"
                            else:
                                if state.Chicken_0.y - state.Car_4.x <= 0.49:
                                    return "UP"
```



# Use attention for smaller skills

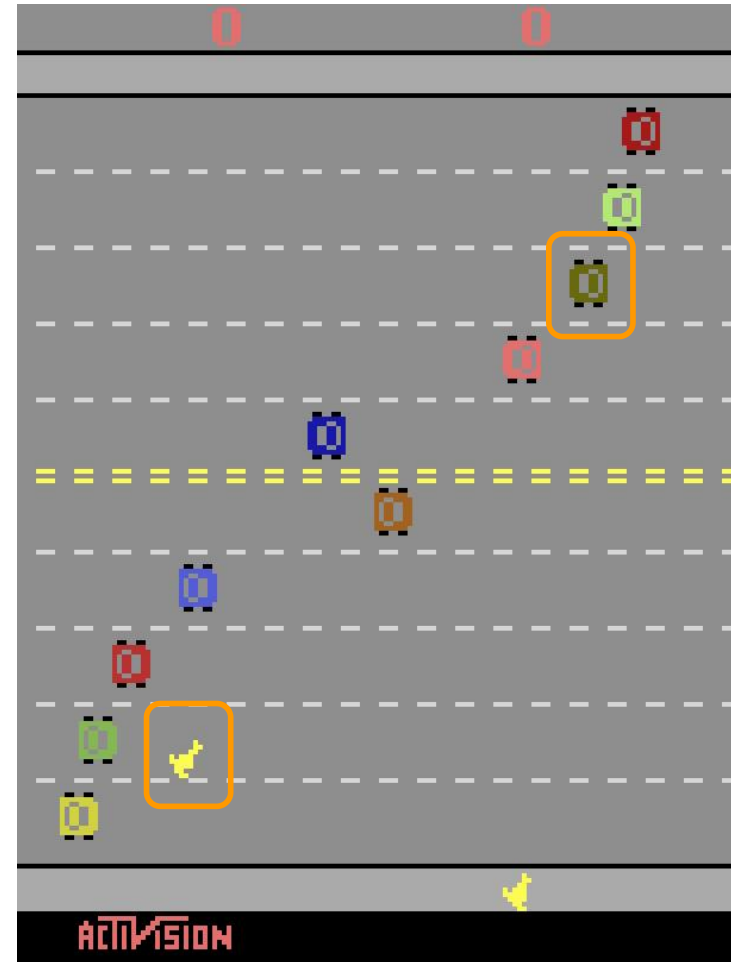
From deep agents – not working



**Let's use object-centric attention!**

# Use object centric attention

From object centric agents





# Transparent policies

Use first order logic

0.57:jump:-closeby(**agent**, **enemy**).

0.32:left[to\_key]:-  $\neg$ has\_key(**agent**), on\_right(**agent**, **key**)

0.30:left[to\_door]:- has\_key(**agent**), on\_left(**door**, **agent**).

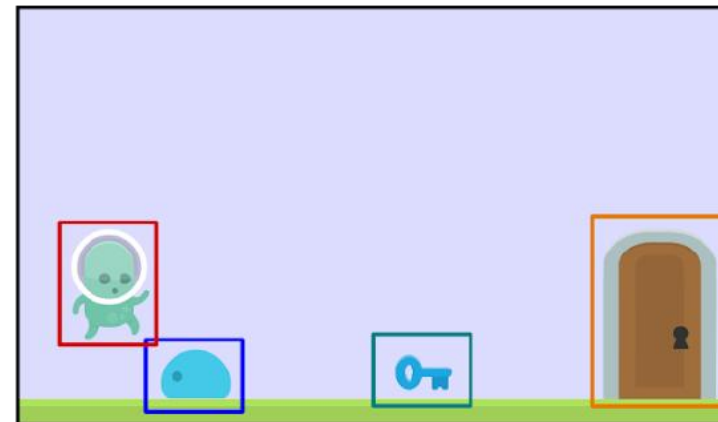
0.29:right[to\_key]:-  $\neg$ has\_key(**agent**), on\_left(**agent**, **key**).

0.56:right[to\_door]:-has\_key(**agent**), on\_left(**agent**, **door**).



# Transparent policies

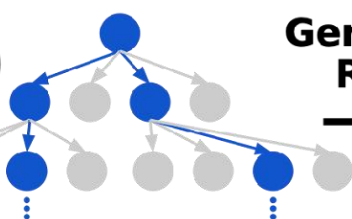
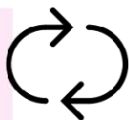
Use first order logic



## Policy Learning

### Neurally-Guided Symbolic Abstraction

Neural Policy

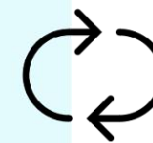


Generated Rules

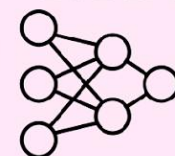


### Differentiable Logic Policy (Actor)

```
0.98:right(1)(agent):-type(01,agent),type(02,key)
                                ¬has_key(01),on_right(02,01).
0.96:right(2)(agent):-type(01,agent),type(02,door)
                                has_key(01),on_right(02,01).
0.65:jump(1)(agent):-type(01,agent),type(02,enemy)
                                closeby(02,01).
⋮
```



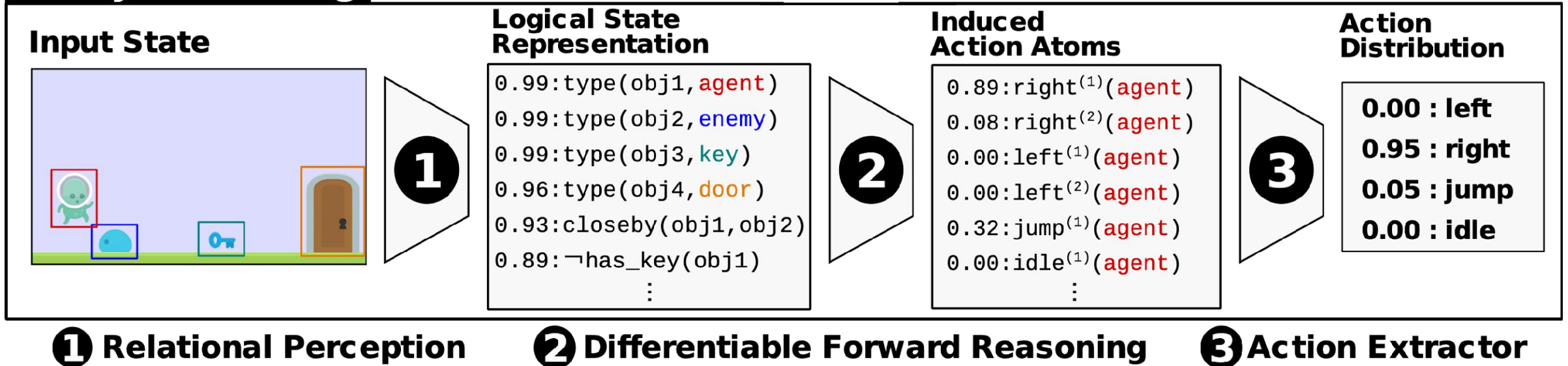
Critic



# Transparent policies

Use first order logic

## Policy Reasoning

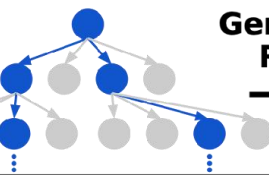


# Transparent policies

Use first order logic

## Policy Learning

Neurally-Guided  
Symbolic Abstraction



Generated  
Rules

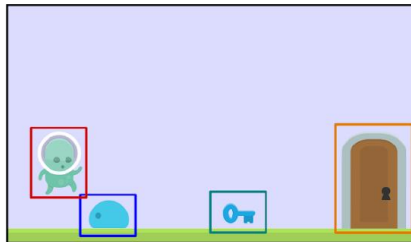
### Differentiable Logic Policy (Actor)

```
0.98:right(1)(agent):-type(01,agent),type(02,key)
                             ¬has_key(01),on_right(02,01).
0.96:right(2)(agent):-type(01,agent),type(02,door)
                             has_key(01),on_right(02,01).
0.65:jump(1)(agent):-type(01,agent),type(02,enemy)
                             closeby(02,01).
⋮
```



## Policy Reasoning

Input State



Logical State  
Representation

```
0.99:type(obj1,agent)
0.99:type(obj2,enemy)
0.99:type(obj3,key)
0.96:type(obj4,door)
0.93:closeby(obj1,obj2)
0.89:¬has_key(obj1)
⋮
```



Induced  
Action Atoms

```
0.89:right(1)(agent)
0.08:right(2)(agent)
0.00:left(1)(agent)
0.00:left(2)(agent)
0.32:jump(1)(agent)
0.00:idle(1)(agent)
⋮
```



Action  
Distribution

```
0.00 : left
0.95 : right
0.05 : jump
0.00 : idle
```

① Relational Perception

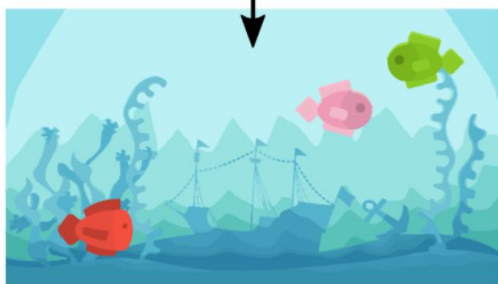
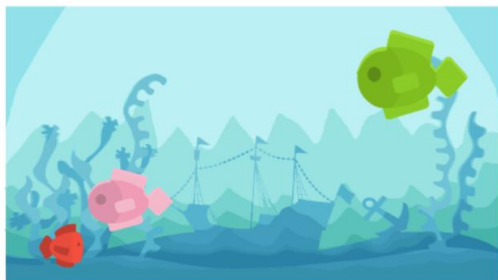
② Differentiable Forward Reasoning

③ Action Extractor

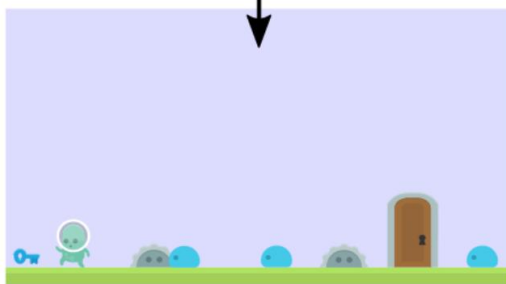
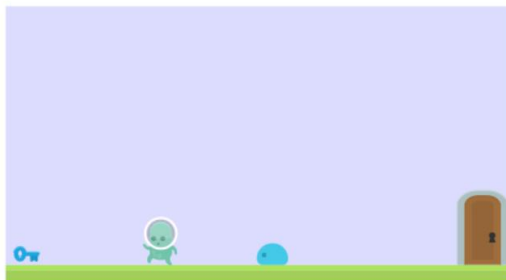


# Transparent policies

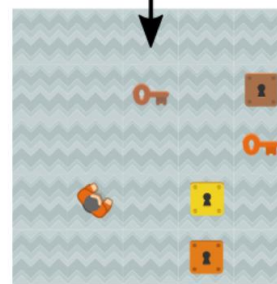
Easily adapts to variations



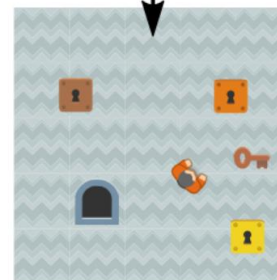
3Fishes-C



GetOut+



Loot-C



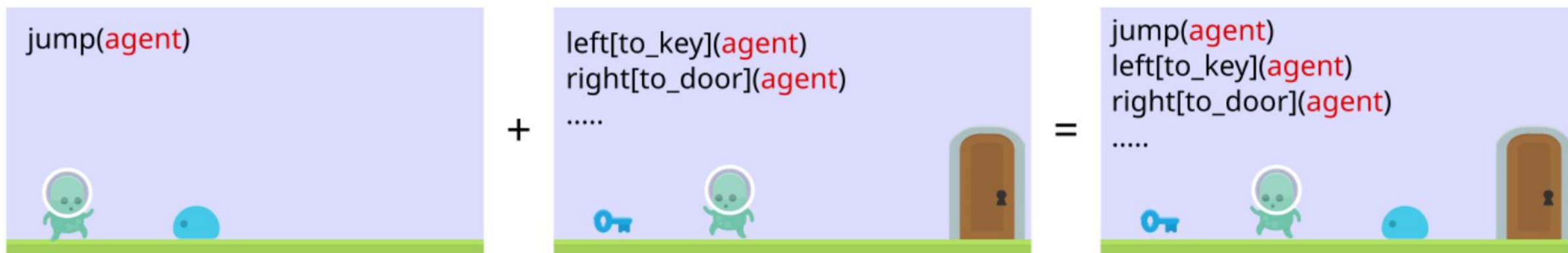
Loot-Hard

| Score ( $\uparrow$ ) | Random           | Neural PPO        | NUDGE                  |
|----------------------|------------------|-------------------|------------------------|
| 3Fishes-C            | -0.64 $\pm$ 0.17 | -0.37 $\pm$ 0.10  | <b>3.26</b> $\pm$ 0.20 |
| GetOut+              | -22.5 $\pm$ 0.41 | -20.88 $\pm$ 0.57 | <b>3.60</b> $\pm$ 2.93 |
| Loot-C               | 0.56 $\pm$ 0.29  | 0.83 $\pm$ 0.49   | <b>5.63</b> $\pm$ 0.33 |
| Loot-Hard            | 1.2 $\pm$ 0.4    | 8.4 $\pm$ 3.2     | 9.2 $\pm$ 2.0          |

# Transparent policies

## Composable Policies

Nudge allows to easily combine skills, learned in continual learning settings.



Nudge agents only need to relearn ordering of the rules, by finetuning the weights.

# Transparent policies

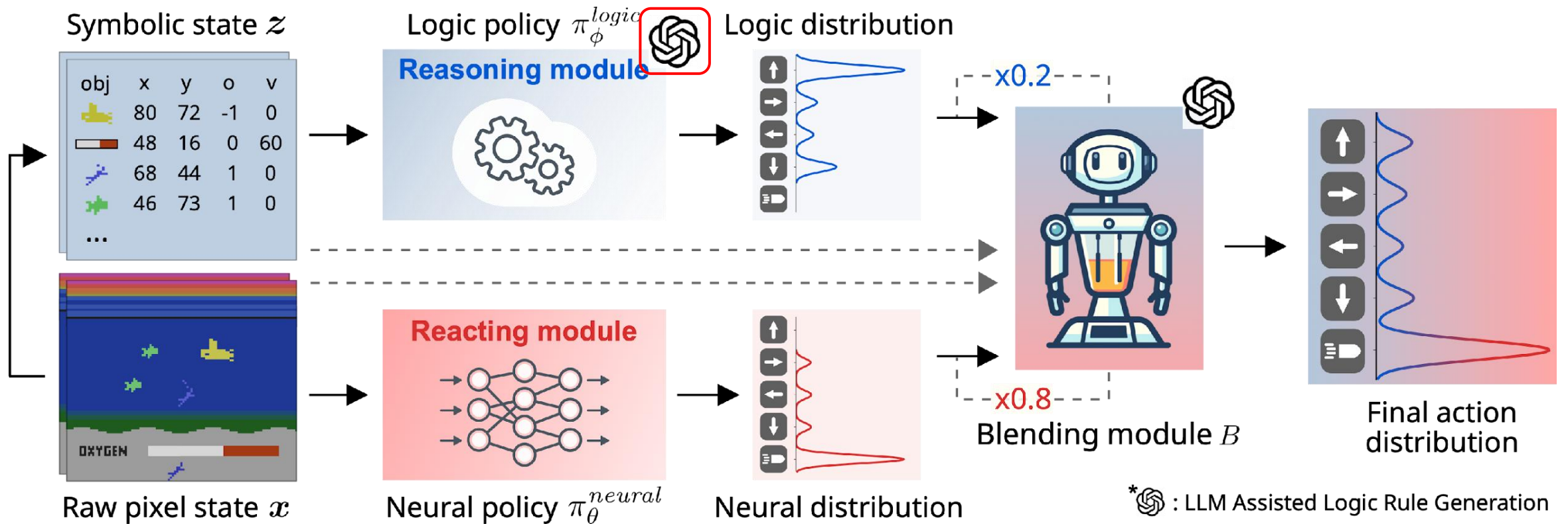
How to handle unknown concepts?



- Predicate invention
- Context + LLMs
- If no context: NNs

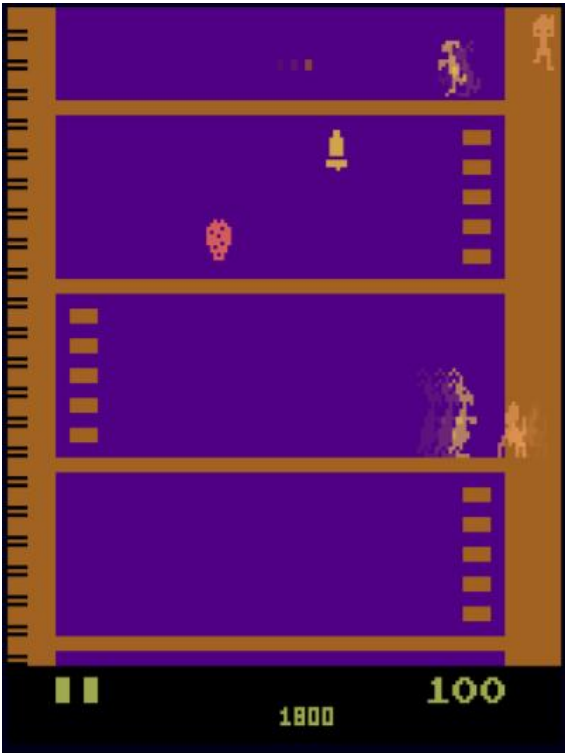
# Overcoming limited knowledge

## Symbols + neural networks



# Overcoming limited knowledge

Use first order logic + NN

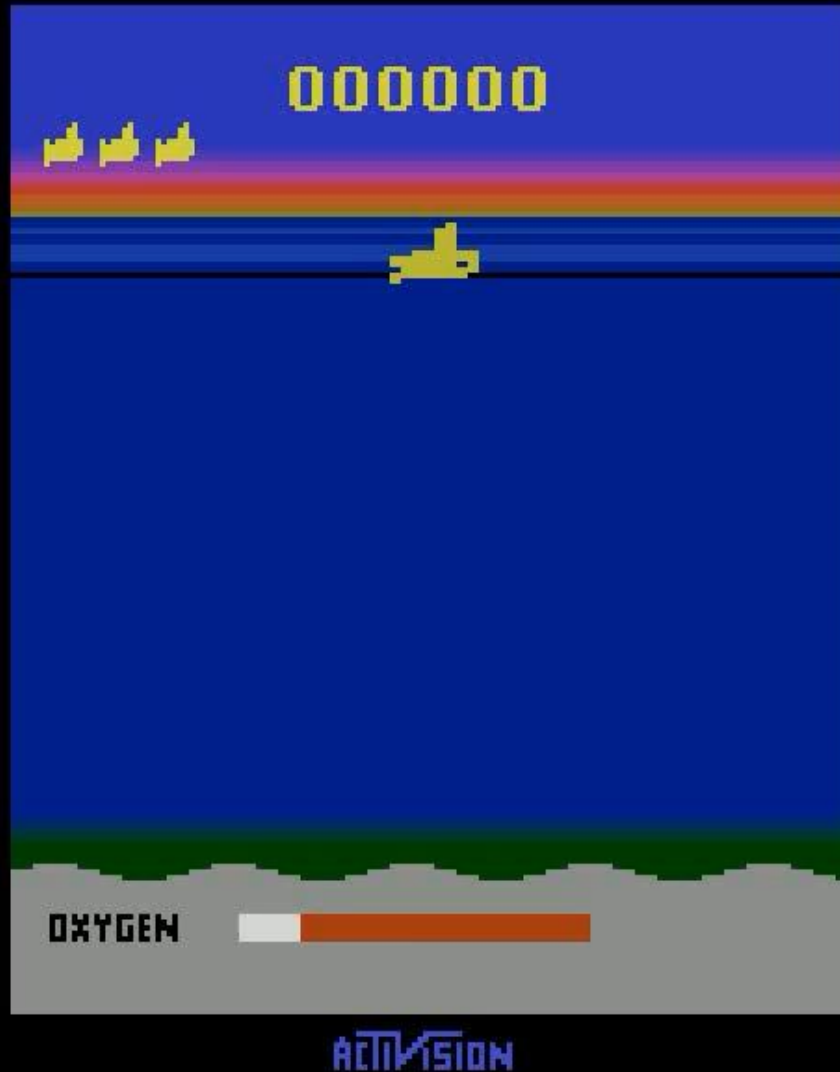


```
% Policy ruleset (Kangaroo)
0.73 up(X):-on_ladder(Player,Ladder),same_floor(Player,Ladder).
0.74 right(X):-left_of(Player,Ladder),same_floor(Player,Ladder).
0.72 left(X):-right_of(Player,Ladder),same_floor(Player,Ladder).
...
% Blending ruleset (Kangaroo)
0.78 neural(X):-closeby(Player,Monkey).
0.51 neural(X):-closeby(Player,Coconut).
0.11 logic(X):-nothing_around(Player).
```





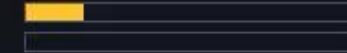
# Interpretable symbolic policies



## SEAQUEST RULE AGENT

### — Game State —

Step: 0  
Score: 0  
Lives: ♥ ♥ ♥  
Oxygen: 11/64  
Divers: 0/6  
Rescues: 0



### — Active Rule —

► hold\_surface\_refill  
→ action: NOOP

### — Matched Rules —

► 8.80 hold\_surface\_refill  
8.70 leave\_surface\_dive  
5.40 dive\_to\_lane2  
0.05 fallback\_noop

### — True Predicates —

always\_true  
at\_surface  
near\_surface  
needs\_divers  
no\_divers  
not\_collision\_risk  
not\_deep\_water  
not\_left\_edge  
not\_nearly\_full  
not\_right\_edge  
not\_right\_zone  
not\_threat\_close  
oxygen\_ok  
oxygen\_return

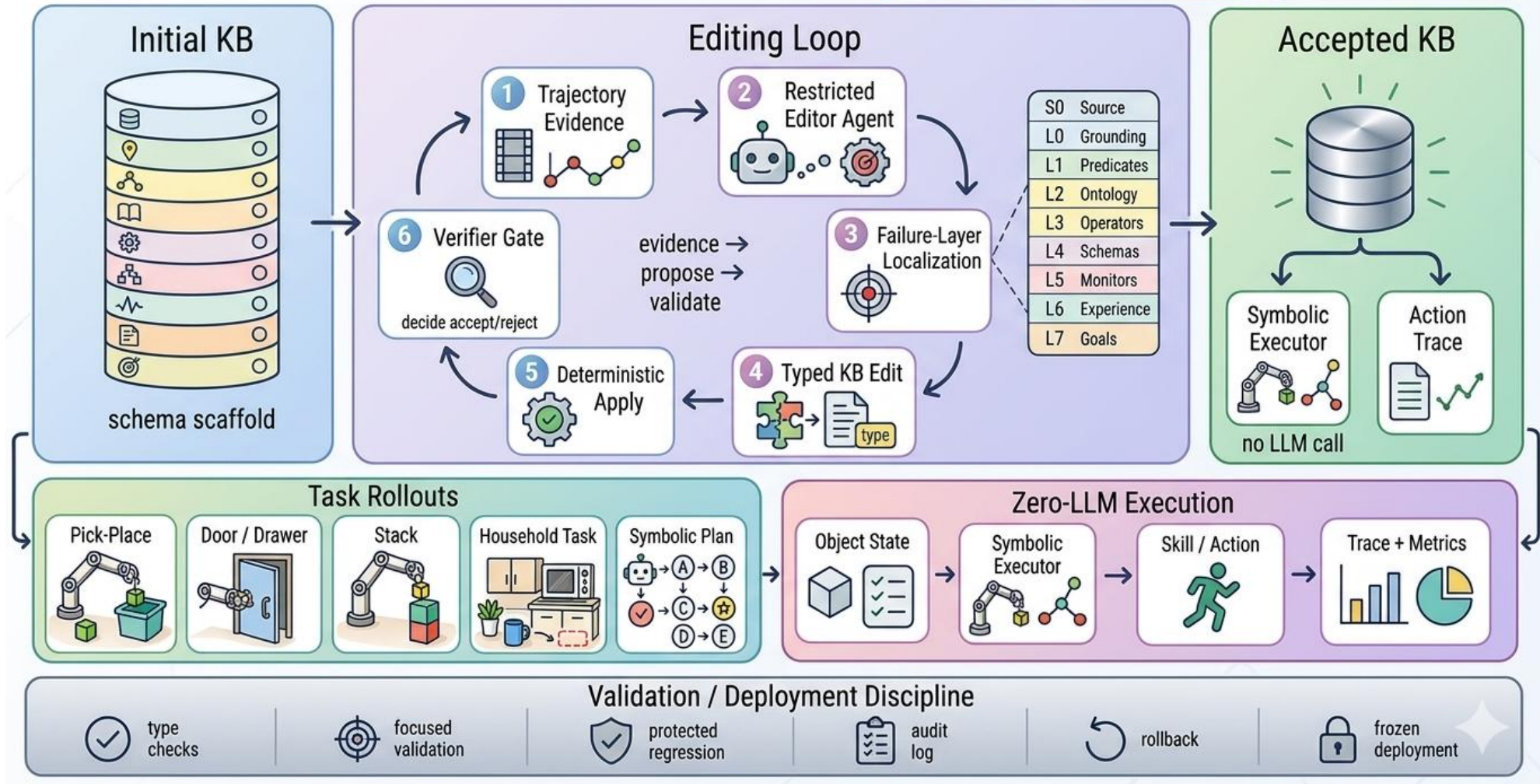
# But the world evolves

## ➤ ILIEA: Policy as Editable Knowledge

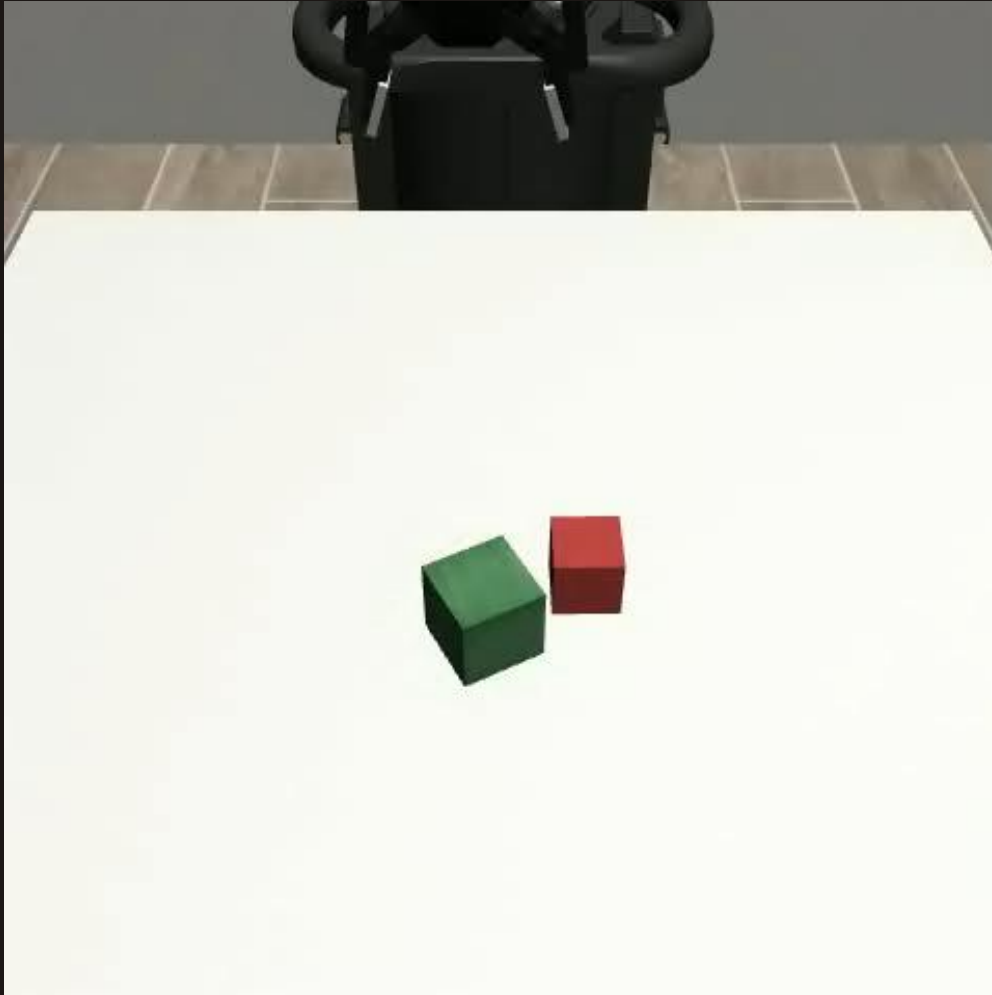
 typed edits

 verifier-gated

 zero-LLM inference



# Symbolic methods in robotics



Stack robot=UR5e STACK\_ITER\_LEVEL=3

t= 0 R= 0.0 stage=pick ??? APPROACH\_A

KNOWLEDGE BASE (rules in priority order)

| prio | name                      | iter  | body ??? | head |
|------|---------------------------|-------|----------|------|
| 10   | done_latch                |       |          | v2   |
| 20   | place_released_on_B_done  |       |          | v2   |
| 30   | place_released_drop_reset |       |          | v2   |
| 40   | place_release_on_b        |       |          | v2   |
| 50   | place_release_aligned     |       |          | v2   |
| 60   | place_descend_b           |       |          | v2   |
| 70   | place_align_b             |       |          | v2   |
| 80   | place_move_to_b           |       |          | v2   |
| 90   | place_lost_grip           |       |          | v2   |
| 100  | pick_held_lifted_to_place |       |          | v2   |
| 108  | pick_held_lift_soft       | iter3 |          | v2   |
| 110  | pick_held_lift            |       |          | v2   |
| 115  | pick_lift_stalled_regrasp | iter2 |          | v2   |
| 120  | pick_grasp_failed_retry   |       |          | v2   |
| 125  | pick_grasp_hold           | iter1 |          | v2   |
| 130  | pick_close_for_lift       |       |          | v2   |
| 140  | pick_grasp_trigger        |       |          | v2   |
| 150  | pick_descend_stuck        |       |          | v2   |
| 160  | pick_descend              |       |          | v2   |
| 170  | pick_align                |       |          | v2   |
| 999  | default_approach          |       |          | v2   |

stacked\_or\_seen ??? DONE  
 stage\_is\_place ??? held\_A ??? gripper\_open ??? o???  
 stage\_is\_place ??? held\_A ??? gripper\_open ??? A???  
 stage\_is\_place ??? on\_B ??? RELEASE  
 stage\_is\_place ??? xy\_aligned\_AB ??? above\_B ??????  
 stage\_is\_place ??? xy\_aligned\_AB ??? above\_B ??????  
 stage\_is\_place ??? held\_A ??? lifted\_A ??? xy\_cl???  
 stage\_is\_place ??? held\_A ??? lifted\_A ??? MOVE???  
 stage\_is\_place ??? held\_A ??? APPROACH\_A  
 stage\_is\_pick ??? held\_A ??? lifted\_A ??? LIFT\_A  
 stage\_is\_pick ??? held\_A ??? lift\_early ??? LIFT???  
 stage\_is\_pick ??? held\_A ??? LIFT\_A  
 stage\_is\_pick ??? lift\_stalled ??? APPROACH\_A  
 stage\_is\_pick ??? gripper\_open ??? held\_A ??? gr???  
 stage\_is\_pick ??? close\_A\_for\_lift ??? grasp\_h???  
 stage\_is\_pick ??? close\_A\_for\_lift ??? LIFT\_A  
 stage\_is\_pick ??? xy\_aligned\_A ??? above\_A ??? g???  
 stage\_is\_pick ??? descend\_stuck ??? GRASP\_A  
 stage\_is\_pick ??? xy\_aligned\_A ??? above\_A ??? D???  
 stage\_is\_pick ??? xy\_close\_A ??? above\_A ??? ALI???

ACTION (continuous)

dx=+0.97 dy=+0.26 dz=-0.73 drx=+0.00  
 dry=+0.00 drz=+0.00 grip=-1.00

PREDICATES TRUE (5)

gripper\_open above\_A stage\_is\_pick  
 grasp\_hold lift\_early



# | IMPORTANT INDUCTIVE BIASES



## Object-centricity

Modeling the world as a compositional structure of discrete entities rather than flat data.



## Hierarchies

Decomposing goals from high-level abstract intentions down to low-level motor actions.



## Relational Reasoning

Explicitly modeling interactions and dependencies between different components of a system.



## Attention

Prioritizing relevant informational features while dynamically suppressing irrelevant context.



## Causality

Moving beyond statistical correlation to understand underlying mechanisms and interventions.



## Locality

Exploiting spatial proximity, gravity, and general world constraints for efficient learning.

---

In essence. . .

A photograph of Rich Sutton, a man with a long white beard and a colorful patterned shirt, smiling and gesturing with his hands. He is standing in front of a presentation slide. The slide features the Reinforcement Learning Conference logo in the top right corner, which consists of a blue cube icon and the text "Reinforcement Learning Conference". The main title of the slide is "THE OaK ARCHITECTURE:" in large white letters on a blue background. Below the title is the subtitle "A Vision of SuperIntelligence from Experience" in blue text. At the bottom of the slide, the speaker's name and affiliations are listed: "Rich Sutton, Keen Technologies, University of Alberta, Amii, Openmind".

Reinforcement Learning Conference

# THE OaK ARCHITECTURE:

A Vision of SuperIntelligence from Experience

**Rich Sutton**, Keen Technologies, University of Alberta, Amii, Openmind

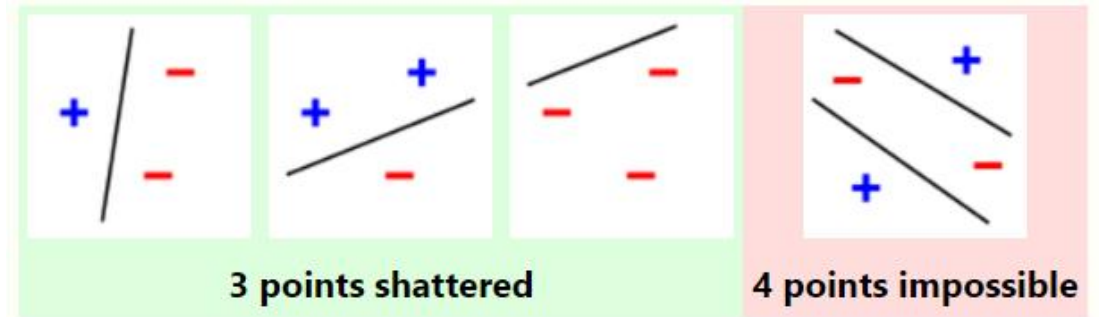


# VC Dimension: Capacity & Generalization

## Measuring Model Complexity

The Vapnik–Chervonenkis (VC) dimension quantifies the **expressive power** of a model class by measuring the maximum number of points it can shatter.

-  **Theoretical Capacity:** Higher VC dimension indicates a model can fit more complex, non-linear patterns.
-  **Generalization Bound:** It provides a mathematical gap between training error and true test error.
-  **Neural Network Paradox:** While deep nets have huge VC dimensions, *Implicit Regularization* (like SGD) helps them avoid overfitting.



# Thanks!





Only symbolic agents can  
learn generalisable policies

CHANGE MY MIND

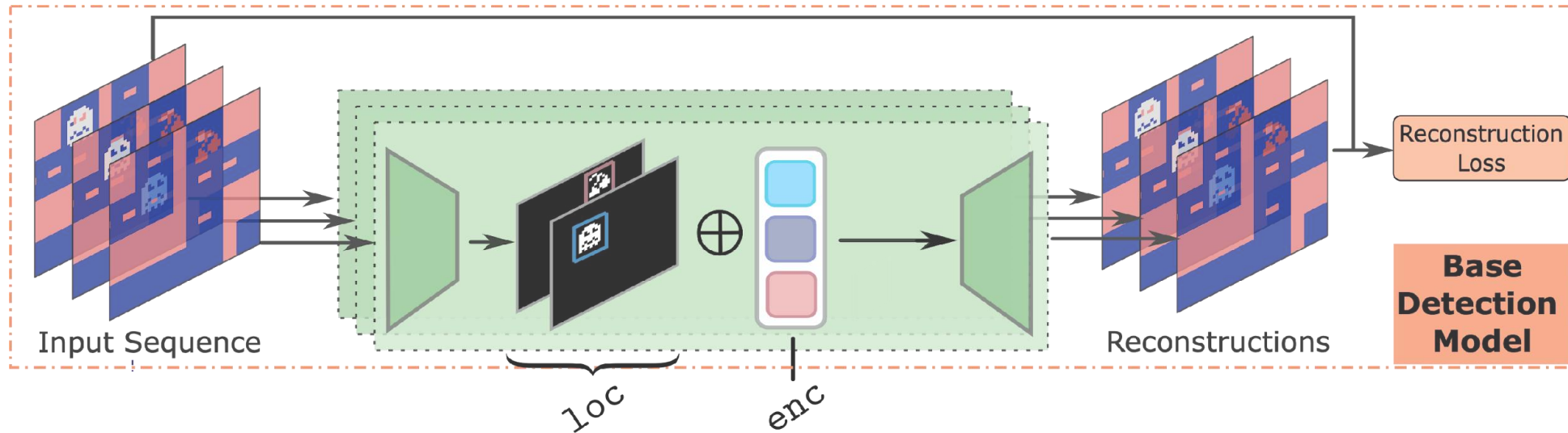
# Any question?





# Object detection needs extra training

Time supervision as human prior



# STORM: 6D pose estimation

